

TEST CASE REPORT

Learning Management System Rajasinghe Central College

Course Code	ICT 3142
Prepared By	Group 04 – Team_404
Module	Software Testing Methods and Tools
Date	May 2026
Team Members	TG/2022/1395 TG/2022/1397 TG/2022/1388 TG/2022/1389

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Requirement 01: Authentication System

Brief: The system shall provide secure login access for Admin, Technical Officer, Teacher, and Student users using valid username and password credentials.

Test Case ID: **AU_001**

Verify login with valid credentials

Test Case Details

Created By	Team_404
Date Created	08-April-2026
Tester Name	Team_404
Date Tested	09-April-2026
Test Case	Pass

Preconditions

S#	Precondition
1	User account exists in the system
2	Internet connection available
3	Login page accessible

Test Data

S#	Data
1	Username = admin01
2	Password = Admin@123

Test Scenario: Verify that users can log in successfully using valid credentials.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Navigate to LMS login page	Login page loads successfully	As Expected	Pass
2	Enter valid username and password	Credentials accepted	As Expected	Pass
3	Click Login button	User redirected to dashboard	As Expected	Pass

Test Case ID: **AU_002**

Verify login with invalid credentials

Test Case Details

Created By	Team_404
Date Created	09-April-2026

Tester Name	Team_404
Date Tested	11-April-2026
Test Case	Fail

Preconditions

S#	Precondition
1	Login page accessible

Test Data

S#	Data
1	Username = wrongUser
2	Password = wrongPass

Test Scenario: Verify the system rejects invalid login credentials.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Enter invalid username and password	Invalid credentials detected	As Expected	Pass
2	Click Login button	Error message displayed	As Expected	Pass
3	Attempt dashboard access	Access denied	As Expected	Pass

Test Case ID: AU_003**Verify unauthorized user cannot access dashboard URL directly****Test Case Details**

Created By	Team_404
Date Created	13-April-2026
Tester Name	Team_404
Date Tested	15-April-2026
Test Case	Pass

Preconditions

S#	Precondition
1	User not logged in

Test Scenario: Ensure unauthorized users cannot directly access protected pages.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Enter dashboard URL manually	System checks authentication	As Expected	Pass

2	Access protected route	Redirected to login page	As Expected	Pass
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Test Case ID: **AU_004**

Verify session timeout after inactivity

Test Case Details

Created By	Team_404
Date Created	17-April-2026
Tester Name	Team_404
Date Tested	19-April-2026
Test Case	Pass

Preconditions

S#	Precondition
1	User is logged in
2	Session timeout set to 15 minutes in system config

Test Scenario: Verify that session expires after user is inactive for the configured duration.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login with valid credentials	User reaches dashboard	As Expected	Pass
2	Leave browser idle for 15 minutes	System monitors inactivity	As Expected	Pass
3	Attempt any action on dashboard	Session expired message shown, redirected to login	As Expected	Pass

Test Case ID: **AU_005**

Verify account lockout after multiple failed login attempts

Test Case Details

Created By	Team_404
Date Created	20-April-2026
Tester Name	Team_404
Date Tested	20-April-2026
Test Case	Fail

Preconditions

S#	Precondition
1	Valid user account exists

2	Lockout policy set to 5 attempts
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Test Data

S#	Data
1	Username = teacher01
2	Password = WrongPass (repeated 5 times)

Test Scenario: Verify that the account is locked after 5 consecutive failed login attempts.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Enter incorrect password 5 times	Each attempt shows error message	As Expected	Pass
2	Attempt 6th login	Account locked message expected	Account locked message not shown – user can still attempt login	Fail
3	Contact admin for unlock	Admin resets account	As Expected	Pass

Test Case ID: AU_006**Verify successful logout functionality****Test Case Details**

Created By	Team_404
Date Created	22-April-2026
Tester Name	Team_404
Date Tested	22-April-2026
Test Case	Pass

Preconditions

S#	Precondition
1	User is logged in as any role

Test Scenario: Verify that users can log out successfully and session is invalidated.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Click Logout button	Logout process initiated	As Expected	Pass
2	User redirected to login page	Login page displayed	As Expected	Pass
3	Press browser Back button	Dashboard not accessible; login page shown	As Expected	Pass

Requirement 02: User Management

Brief: Admin and Technical Officer should be able to manage users in the LMS.

Test Case ID: **UM_001**

Verify Admin can add a new student account

Test Case Details

Created By	Team_404
Date Created	22-April-2026
Tester Name	Team_404
Date Tested	22-April-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Admin logged into system
2	Add User form accessible

Test Data

S#	Data
1	Student Name = Nimal Perera
2	Username = st1001
3	Role = Student

Test Scenario: Verify Admin can create a student account successfully.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Navigate to User Management page	User page loads	As Expected	Pass
2	Fill student details	Form accepts data	As Expected	Pass
3	Click Create User	Student account created successfully	As Expected	Pass

Test Case ID: **UM_002**

Verify system prevents duplicate usernames

Test Case Details

Created By	Team_404
Date Created	22-April-2026
Tester Name	Team_404

Date Tested	23-April-2026
Test Case	Fail

Preconditions

S#	Precondition
1	Existing username already stored in database

Test Scenario: Verify duplicate usernames are not allowed.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Enter existing username	System validates uniqueness	As Expected	Pass
2	Submit registration form	Error message displayed	As Expected	Pass

Test Case ID: UM_003**Verify Technical Officer can remove a user**

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Technical Officer	Dashboard loads	As Expected	Pass
2	Navigate to user list	User records displayed	As Expected	Pass
3	Click Delete on selected user	Confirmation dialog shown	As Expected	Pass
4	Confirm deletion	User removed successfully	As Expected	Pass

Test Case ID: UM_004**Verify Teacher cannot access user management module**

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Teacher	Teacher dashboard loads	As Expected	Pass
2	Attempt to access User Management page	Access denied message shown	As Expected	Pass

Test Case ID: UM_005**Verify Admin can edit an existing user's details****Test Case Details**

Created By	Team_404
Date Created	22-April-2026

Tester Name	Team_404
Date Tested	23-April-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Admin logged in
2	User record exists

Test Data

S#	Data
1	User = st1001
2	New Email = nimal.p@rcc.edu.lk

Test Scenario: Verify Admin can update user information.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Open User Management and search user	User record found	As Expected	Pass
2	Click Edit and update email field	Field accepts new value	As Expected	Pass
3	Click Save	User record updated with new email	As Expected	Pass

Test Case ID: UM_006**Verify Admin can reset a user password****Test Case Details**

Created By	Team_404
Date Created	22-April-2026
Tester Name	Team_404
Date Tested	24-April-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Admin logged in
2	Target user account exists

Test Scenario: Verify Admin can reset a user's password and user receives notification.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Select user from User Management	User profile opens	As Expected	Pass
2	Click Reset Password	Temporary password generated	As Expected	Pass
3	Confirm reset	User notified via system message	As Expected	Pass
4	User logs in with temporary password	Login successful	As Expected	Pass

Requirement 03: Class Management

Brief: Admin and Technical Officer should be able to manage classes.

Test Case ID: **CM_001**

Verify Admin can create a class successfully

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Navigate to Class Management	Page loads successfully	As Expected	Pass
2	Enter class details (Name: Grade 10A)	Valid data accepted	As Expected	
3	Click Save	Class created successfully	As Expected	

Test Case ID: **CM_002**

Verify system rejects empty class name

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Leave class name empty	Validation triggered	As Expected	Pass
2	Submit form	Error message displayed	As Expected	

Test Case ID: **CM_003**

Verify Admin can assign students to classes

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Select class	Student assignment page opens	As Expected	Pass
2	Select students	Students selected successfully	As Expected	
3	Save assignments	Students assigned to class	As Expected	

Test Case ID: **CM_004**

Verify Admin can delete a class

Test Case Details

Created By	Team_404
Date Created	22-April-2026
Tester Name	Team_404

Date Tested	24-April-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Admin logged in
2	Class with no active students exists

Test Scenario: Verify Admin can delete an empty class.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Open Class Management	Class list displayed	As Expected	Pass
2	Select class to delete	Confirmation dialog shown	As Expected	Pass
3	Confirm deletion	Class removed from system	As Expected	Pass

Test Case ID: CM_005**Verify system prevents deletion of class with enrolled students****Test Case Details**

Created By	Team_404
Date Created	24-April-2026
Tester Name	Team_404
Date Tested	24-April-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Admin logged in
2	Class with enrolled students exists

Test Scenario: Verify that a class cannot be deleted if it has enrolled students.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Attempt to delete active class	System checks enrolment	As Expected	Pass
2	Confirm deletion attempt	Error: Cannot delete class with enrolled students	As Expected	Pass

Requirement 04: Subject Management

Brief: The system shall support creation and assignment of subjects.

Test Case ID: SM_001

Verify Admin can create subjects

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Open Subject Management page	Page loads successfully	As Expected	Pass
2	Enter subject details	Subject data accepted	As Expected	Pass
3	Save subject	Subject added successfully	As Expected	Pass

Test Case ID: SM_002

Verify teacher assignment to subjects

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Select subject	Subject selected	As Expected	Pass
2	Assign teacher	Teacher assigned successfully	As Expected	Pass

Test Case ID: SM_003

Verify Sinhala and English medium subjects are supported

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Create Sinhala medium subject	Subject created successfully	As Expected	Pass
2	Create English medium subject	Subject created successfully	As Expected	Pass

Test Case ID: SM_004

Verify subject can be assigned to multiple classes

Test Case Details

Created By	Team_404
Date Created	24-April-2026
Tester Name	Team_404

Date Tested	24-April-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Multiple classes exist
2	Subject exists in the system

Test Scenario: Verify a single subject can be linked to more than one class.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Open subject and click Assign to Class	Assignment dialog opens	As Expected	Pass
2	Select Grade 10A	Class added to subject	As Expected	Pass
3	Select Grade 10B additionally	Second class added	As Expected	Pass
4	Save changes	Subject linked to both classes	As Expected	Pass

Requirement 05: Marks Management

Brief: Teachers should be able to upload student marks and students should view marks.

Test Case ID: MM_001

Verify teacher uploads marks successfully

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Teacher	Dashboard loads	As Expected	Pass
2	Navigate to Marks Management	Marks page displayed	As Expected	
3	Enter marks and save	Marks stored successfully	As Expected	Pass

Test Case ID: MM_002

Verify system rejects marks above valid range

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Enter marks greater than 100	Validation triggered	As Expected	Pass
2	Save marks	Error message displayed	As Expected	Pass

Test Case ID: MM_003

Verify students can view marks

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Student	Dashboard loads	As Expected	Pass
2	Open Marks section	Marks displayed correctly	As Expected	

Test Case ID: MM_004

Verify students cannot edit marks

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Student opens marks page	Marks visible	As Expected	Pass
2	Attempt to edit marks	Edit option unavailable	As Expected	

Test Case ID: MM_005

Verify marks are correctly calculated as percentage

Test Case Details

Created By	Team_404
Date Created	24-April-2026
Tester Name	Team_404
Date Tested	25-April-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Teacher logged in
2	Student marks entry form accessible

Test Data

S#	Data
1	Marks Obtained = 75
2	Total Marks = 100
3	Expected Percentage = 75%

Test Scenario: Verify the system computes and displays percentage correctly.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Enter marks obtained and total marks	Fields accept values	As Expected	Pass
2	Save marks	System calculates percentage	As Expected	Pass
3	Student views marks	Percentage shown as 75%	As Expected	Pass

Test Case ID: MM_006

Verify marks can be updated after initial entry

Test Case Details

Created By	Team_404
Date Created	24-April-2026
Tester Name	Team_404
Date Tested	25-April-2026
Test Case	Fail

Preconditions

S#	Precondition
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1	Teacher logged in
2	Marks already saved for student

Test Scenario: Verify teacher can update previously entered marks.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Open existing marks record	Record displayed	As Expected	Pass
2	Modify marks value	Field allows edit	As Expected	Pass
3	Save updated marks	Updated value saved expected – system shows old value after refresh	Updated marks not persisted after page refresh	Fail

Requirement 06: Study Material Management

Brief: Teachers should upload study materials and students should access them.

Test Case ID: STM_001

Verify teacher uploads PDF successfully

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Open Upload Material page	Upload form loads	As Expected	Pass
2	Select valid PDF file	File accepted	As Expected	
3	Click Upload	Material uploaded successfully	As Expected	Pass

Test Case ID: STM_002

Verify invalid file upload rejection

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Select unsupported file type (.exe)	Validation triggered	As Expected	Pass
2	Upload file	Error message displayed	As Expected	

Test Case ID: STM_003

Verify YouTube link upload functionality

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Enter valid YouTube link	Link accepted	As Expected	Pass
2	Save material	Link stored successfully	As Expected	

Test Case ID: STM_004

Verify students can download study materials

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Student opens material page	Materials displayed	As Expected	Pass
2	Click Download on PDF	File downloaded successfully	As Expected	

Test Case ID: STM_005**Verify file size limit is enforced on upload****Test Case Details**

Created By	Team_404
Date Created	24-April-2026
Tester Name	Team_404
Date Tested	24-April-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Teacher logged in
2	File size limit set to 20MB in system settings

Test Data

S#	Data
1	File = large_document.pdf (25MB)

Test Scenario: Verify the system rejects files exceeding the maximum size limit.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Select file exceeding 20MB	File selected locally	As Expected	Pass
2	Attempt upload	System checks file size	As Expected	Pass
3	Observe result	Error: File exceeds 20MB limit displayed	As Expected	Pass

Test Case ID: STM_006**Verify teacher can delete uploaded material****Test Case Details**

Created By	Team_404
Date Created	26-April-2026
Tester Name	Team_404
Date Tested	26-April-2026
Test Case	Pass

Preconditions

S#	Precondition
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1	Teacher logged in
2	At least one study material uploaded

Test Scenario: Verify teacher can remove their own uploaded study material.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Open Study Materials page	Material list shown	As Expected	Pass
2	Click Delete on a material	Confirmation dialog appears	As Expected	Pass
3	Confirm delete	Material removed from list	As Expected	Pass

Requirement 07: Report Generation

Brief: The LMS shall generate student and school academic reports.

Test Case ID: RP_001

Verify student performance report generation

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Open Reports page	Reports page loads	As Expected	Pass
2	Select student	Student data retrieved	As Expected	Pass
3	Generate report	Report generated successfully	As Expected	Pass

Test Case ID: RP_002

Verify class-wise report generation

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Select class	Class selected	As Expected	Pass
2	Click Generate	Class report generated	As Expected	Pass

Test Case ID: RP_003

Verify unauthorized report access restriction

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Student	Student dashboard loads	As Expected	Pass
2	Attempt to access school-wide report	Access denied	As Expected	Pass

Test Case ID: RP_004

Verify report can be exported as PDF

Test Case Details

Created By	Team_404
Date Created	26-April-2026
Tester Name	Team_404
Date Tested	26-April-2026

Test Case	Pass
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Preconditions

S#	Precondition
1	Admin logged in
2	At least one student with marks exists

Test Scenario: Verify generated report can be downloaded as a PDF file.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Generate a student performance report	Report displayed on screen	As Expected	Pass
2	Click Export as PDF	PDF download initiated	As Expected	Pass
3	Open downloaded file	Report content displayed correctly in PDF	As Expected	Pass

Requirement 08: Student Profile Management

Brief: Students should be able to view their profiles.

Test Case ID: SP_001

Verify student profile viewing

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Student	Dashboard loads	As Expected	Pass
2	Open Profile page	Profile information displayed	As Expected	Pass

Test Case ID: SP_002

Verify Admin can update student profile details

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Admin	Dashboard loads	As Expected	Pass
2	Edit student information	Form accepts updates	As Expected	Pass
3	Save changes	Student profile updated	As Expected	Pass

Test Case ID: SP_003

Verify student profile displays correct class and subjects

Test Case Details

Created By	Team_404
Date Created	29-April-2026
Tester Name	Team_404
Date Tested	29-April-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Student assigned to a class with subjects

Test Scenario: Verify profile accurately reflects class enrolment and subject list.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Student and open Profile	Profile page loads	As Expected	Pass

2	Check Class field	Correct class shown (e.g. Grade 10A)	As Expected	Pass
3	Check Subjects list	All enrolled subjects listed correctly	As Expected	Pass

Requirement 09: School Website Management

Brief: Technical Officer should manage public website content.

Test Case ID: WEB_001

Verify Technical Officer updates website content

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Technical Officer	Dashboard loads	As Expected	Pass
2	Navigate to Website Management	Website editor displayed	As Expected	Pass
3	Update homepage content	Changes saved successfully	As Expected	Pass

Test Case ID: WEB_002

Verify public users can view website pages

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Open public website	Homepage loads	As Expected	Pass
2	Navigate to Events page	Events displayed correctly	As Expected	Pass

Test Case ID: WEB_003

Verify Technical Officer can add school news/announcements

Test Case Details

Created By	Team_404
Date Created	29-April-2026
Tester Name	Team_404
Date Tested	29-April-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Technical Officer logged in
2	News/Announcements section accessible

Test Scenario: Verify announcements can be created and appear on public site.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
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1	Navigate to Announcements section	Announcement form displayed	As Expected	Pass
2	Enter announcement title and content	Form accepts data	As Expected	Pass
3	Publish announcement	Announcement saved	As Expected	Pass
4	View public website	Announcement visible on homepage	As Expected	Pass

Requirement 10: Role-Based Access Control

Brief: The system shall restrict access according to user roles.

Test Case ID: RBAC_001

Verify Teacher cannot manage users

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Teacher	Teacher dashboard loads	As Expected	Pass
2	Attempt to open User Management	Access denied	As Expected	Pass

Test Case ID: RBAC_002

Verify Student cannot upload marks

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Student	Student dashboard loads	As Expected	Pass
2	Attempt to access Marks Upload page	Permission denied	As Expected	Pass

Test Case ID: RBAC_003

Verify only Admin and Technical Officer can create users

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Teacher	Dashboard loads	As Expected	Pass
2	Attempt user creation	Access denied	As Expected	Pass
3	Login as Admin	Dashboard loads	As Expected	Pass
4	Create user	User created successfully	As Expected	Pass

Test Case ID: RBAC_004

Verify Student cannot access Report Generation module

Test Case Details

Created By	Team_404
Date Created	01-May-2026
Tester Name	Team_404

Date Tested	01-May-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Student account exists and is logged in

Test Scenario: Verify student is blocked from the admin report generation panel.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Student	Student dashboard loads	As Expected	Pass
2	Manually navigate to /admin/reports	System checks role	As Expected	Pass
3	Observe result	Access denied page displayed	As Expected	Pass

Requirement 11: Notification System

Brief: The LMS shall send in-system notifications to users for key events such as new marks, study material uploads, and announcements.

Test Case ID: **NT_001**

Verify student receives notification when marks are uploaded

Test Case Details

Created By	Team_404
Date Created	02-May-2026
Tester Name	Team_404
Date Tested	02-May-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Student account exists
2	Teacher has uploaded marks for the student's class

Test Scenario: Verify student gets an in-system notification upon marks upload.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Teacher uploads marks for Grade 10A	Marks saved successfully	As Expected	Pass
2	Student in Grade 10A logs in	Dashboard loads	As Expected	Pass
3	Check notification bell	New notification: 'Your marks have been updated'	As Expected	Pass
4	Click notification	Student redirected to marks page	As Expected	Pass

Test Case ID: **NT_002**

Verify student receives notification on new study material upload

Test Case Details

Created By	Team_404
Date Created	08-May-2026
Tester Name	Team_404
Date Tested	08-May-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Teacher logged in
2	Student enrolled in relevant class

Test Scenario: Verify notification is sent when teacher uploads new material.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Teacher uploads new PDF for Mathematics	Material uploaded	As Expected	Pass
2	Student logs in	Dashboard loads	As Expected	Pass
3	Check notifications	Notification: 'New study material available for Mathematics'	As Expected	Pass

Test Case ID: NT_003

Verify Admin can send broadcast announcements to all users

Test Case Details

Created By	Team_404
Date Created	08-May-2026
Tester Name	Team_404
Date Tested	08-May-2026
Test Case	Fail

Preconditions

S#	Precondition
1	Admin logged in
2	Multiple user accounts exist

Test Scenario: Verify Admin can broadcast a notification to all system users.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Admin navigates to Announcements	Announcement panel opens	As Expected	Pass
2	Compose and send broadcast message	Message sent to all users expected	Notification delivered only to students; Teachers not receiving broadcast	Fail
3	Various users log in and check notifications	All users see the broadcast	Teachers did not receive notification	Fail

Test Case ID: NT_004**Verify notifications are marked as read when clicked****Test Case Details**

Created By	Team_404
Date Created	08-May-2026
Tester Name	Team_404
Date Tested	08-May-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Unread notification exists for a user

Test Scenario: Verify notification status changes to read upon clicking.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login and open notification panel	Unread notification shown in bold	As Expected	Pass
2	Click on the notification	Redirected to relevant page	As Expected	Pass
3	Return to notification panel	Notification marked as read (no longer bold)	As Expected	Pass

Requirement 12: Attendance Management

Brief: Teachers should be able to record and manage daily student attendance. Students and Admin should be able to view attendance records.

Test Case ID: **AT_001**

Verify teacher can mark student attendance

Test Case Details

Created By	Team_404
Date Created	08-May-2026
Tester Name	Team_404
Date Tested	08-May-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Teacher logged in
2	Class with enrolled students exists
3	Attendance module accessible

Test Data

S#	Data
1	Class = Grade 10A
2	Date = 08-May-2026
3	Student = Nimal Perera – Present

Test Scenario: Verify teacher can submit daily attendance for the class.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Navigate to Attendance module	Attendance page loads	As Expected	Pass
2	Select class and date	Student list displayed	As Expected	Pass
3	Mark each student Present/Absent	Checkboxes respond correctly	As Expected	Pass
4	Click Submit Attendance	Attendance saved successfully	As Expected	Pass

Test Case ID: **AT_002**

Verify system prevents duplicate attendance entry for same date

Test Case Details

Created By	Team_404
Date Created	08-May-2026
Tester Name	Team_404
Date Tested	08-May-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Attendance already submitted for Grade 10A on 08-May-2026

Test Scenario: Verify teacher cannot submit attendance twice for the same date.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Attempt to open attendance form for same class and date	System checks existing record	As Expected	Pass
2	Observe result	Warning: Attendance already submitted for this date	As Expected	Pass

Test Case ID: AT_003**Verify student can view own attendance record****Test Case Details**

Created By	Team_404
Date Created	08-May-2026
Tester Name	Team_404
Date Tested	08-May-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Attendance records exist for the student
2	Student logged in

Test Scenario: Verify student can view their monthly attendance summary.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Student and navigate to Attendance	Attendance page loads	As Expected	Pass
2	View monthly summary	Days Present, Absent, and Percentage shown	As Expected	Pass

Test Case ID: AT_004**Verify Admin can generate attendance report for a class****Test Case Details**

Created By	Team_404
Date Created	08-May-2026
Tester Name	Team_404
Date Tested	08-May-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Admin logged in
2	Attendance data exists for class

Test Scenario: Verify Admin can pull a full attendance report for a selected class.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Go to Reports and select Attendance Report	Report options shown	As Expected	Pass
2	Select Grade 10A and date range	Data retrieved	As Expected	Pass
3	Click Generate	Class attendance report generated with all students	As Expected	Pass

Requirement 13: Timetable Management

Brief: Admin should be able to create and assign class timetables. Teachers and Students should be able to view their respective timetables.

Test Case ID: TT_001

Verify Admin can create a timetable for a class

Test Case Details

Created By	Team_404
Date Created	08-May-2026
Tester Name	Team_404
Date Tested	08-May-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Admin logged in
2	Class and subjects exist
3	Teachers assigned to subjects

Test Data

S#	Data
1	Class = Grade 10A
2	Monday 8:00AM = Mathematics – Mr. Dilshan
3	Monday 9:00AM = Science – Ms. Kavindi

Test Scenario: Verify Admin can build and save a weekly timetable.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Navigate to Timetable Management	Timetable builder loads	As Expected	Pass
2	Select class and add period slots	Subjects and teachers assigned per slot	As Expected	Pass
3	Save timetable	Timetable saved for Grade 10A	As Expected	Pass

Test Case ID: TT_002

Verify student can view class timetable

Test Case Details

Created By	Team_404
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Date Created	08-May-2026
Tester Name	Team_404
Date Tested	08-May-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Student enrolled in a class with a timetable

Test Scenario: Verify student can access their weekly timetable.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Student	Dashboard loads	As Expected	Pass
2	Navigate to Timetable	Weekly timetable displayed with subjects and teachers	As Expected	Pass

Test Case ID: TT_003**Verify teacher can view personal teaching timetable****Test Case Details**

Created By	Team_404
Date Created	08-May-2026
Tester Name	Team_404
Date Tested	08-May-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Teacher assigned to subjects across classes

Test Scenario: Verify teacher timetable shows all their scheduled periods.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Login as Teacher	Dashboard loads	As Expected	Pass
2	Open My Timetable	Teacher's scheduled periods across all classes shown	As Expected	Pass

Test Case ID: TT_004

Verify system detects teacher scheduling conflict

Test Case Details

Created By	Team_404
Date Created	08-May-2026
Tester Name	Team_404
Date Tested	08-May-2026
Test Case	Fail

Preconditions

S#	Precondition
1	Teacher already assigned to Grade 10A on Monday 8:00AM

Test Scenario: Verify system prevents assigning the same teacher to two classes at the same time.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Assign same teacher to Grade 11B on Monday 8:00AM	System checks for conflict	As Expected	Pass
2	Observe result	Conflict warning expected – no warning displayed; timetable saved with conflict	Scheduling conflict not detected by system	Fail

Requirement 14: Password & Security Management

Brief: Users should be able to change their passwords securely. The system must enforce password strength policies.

Test Case ID: **PS_001**

Verify user can change their own password

Test Case Details

Created By	Team_404
Date Created	08-May-2026
Tester Name	Team_404
Date Tested	08-May-2026
Test Case	Pass

Preconditions

S#	Precondition
1	User logged in
2	Change Password option accessible from profile

Test Data

S#	Data
1	Current Password = Admin@123
2	New Password = NewPass@2026
3	Confirm Password = NewPass@2026

Test Scenario: Verify user can update their password through the profile settings.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Open Profile > Change Password	Password change form displayed	As Expected	Pass
2	Enter current and new password	Fields accept input	As Expected	Pass
3	Submit form	Password updated successfully, user prompted to log in again	As Expected	Pass

Test Case ID: **PS_002**

Verify system rejects weak passwords

Test Case Details

Created By	Team_404
Date Created	08-May-2026

Tester Name	Team_404
Date Tested	08-May-2026
Test Case	Pass

Preconditions

S#	Precondition
1	Password policy: min 8 chars, 1 uppercase, 1 number, 1 special char

Test Data

S#	Data
1	Weak Password = 12345
2	Weak Password = password
3	Strong Password = Str0ng@Pass

Test Scenario: Verify weak passwords are rejected during change/registration.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Enter '12345' as new password	Validation triggered	As Expected	Pass
2	Enter 'password' as new password	Validation triggered	As Expected	Pass
3	Enter 'Str0ng@Pass'	Password accepted	As Expected	Pass

Test Case ID: PS_003**Verify password and confirm password must match****Test Case Details**

Created By	Team_404
Date Created	08-May-2026
Tester Name	Team_404
Date Tested	08-May-2026
Test Case	Pass

Preconditions

S#	Precondition
1	User on Change Password form

Test Data

S#	Data
1	New Password = NewPass@2026

2	Confirm Password = DifferentPass@2026
---	---------------------------------------

Test Scenario: Verify mismatched confirm password is caught by validation.

Step #	Step Details	Expected Results	Actual Results	Pass/Fail
1	Enter mismatched passwords	Fields accept input	As Expected	Pass
2	Click Submit	Error: Passwords do not match displayed	As Expected	

Requirement 15: Bug / Defect Log

This section documents all defects identified during test execution, including severity, status, and resolution details.

Bug ID	Test Case	Module	Severity	Description	Status	Remarks
BUG_001	AU_002	Authentication	High	Error message not displayed for invalid login on mobile browser	Open	Reproduced on Safari iOS 17; Chrome unaffected. Needs front-end fix.
BUG_002	AU_005	Authentication	Medium	Account not locked after 5 failed attempts	Open	Lockout policy not enforced. Backend validation missing.
BUG_003	MM_006	Marks Mgmt	High	Updated marks not saved after page refresh	Open	Data not persisting – suspected API PUT endpoint issue.
BUG_004	NT_003	Notifications	Medium	Broadcast notification not reaching Teacher role	In Progress	Role filter bug in notification dispatch service.
BUG_005	TT_004	Timetable	Medium	Teacher scheduling conflict not detected	Open	Conflict check logic missing in timetable save handler.
BUG_006	UM_002	User Mgmt	Low	Duplicate username error message lacks clear guidance	Closed	UI message updated to: 'This username is already taken. Please choose another.'
BUG_007	STM_002	Study Material	Low	Unsupported file error disappears too quickly (under 2 sec)	Closed	Toast notification duration extended to 5 seconds.
BUG_008	RP_004	Reports	Low	PDF export adds blank page at end of report	Open	LaTeX/PDF rendering issue. Under investigation.

Requirement 16: Test Summary & Metrics

This section presents a high-level summary of all test execution results across all 14 requirements.

Overall Test Execution Summary

52 Total Test Cases	45 Test Cases Passed
7 Test Cases Failed	86.5% Pass Rate

Defect Summary

8 Total Bugs Found	2 High Severity
3 Medium Severity	3 Low Severity

Defect Status

5 Open	1 In Progress
2 Closed / Resolved	2 Regression Pending

Results by Requirement Module

Module	Total	Passed	Failed	Pass Rate
Authentication System	6	5	1	83.3%
User Management	6	6	0	100%
Class Management	5	5	0	100%
Subject Management	4	4	0	100%
Marks Management	6	5	1	83.3%
Study Material Management	6	6	0	100%
Report Generation	4	4	0	100%

Student Profile Management	3	3	0	100%
School Website Management	3	3	0	100%
Role-Based Access Control	4	4	0	100%
Notification System (NEW)	4	3	1	75.0%
Attendance Management (NEW)	4	4	0	100%
Timetable Management (NEW)	4	3	1	75.0%
Password & Security (NEW)	3	3	0	100%
TOTAL	52	45	7	86.5%

Requirement 17: Conclusion

This extended test case report covers all functional and security requirements of the Learning Management System developed for Rajasinghe Central College under course ICT 3183. A total of 52 test cases were executed across 14 requirement areas, achieving an overall pass rate of 86.5%.

Four new requirement modules were introduced in this version — Notification System, Attendance Management, Timetable Management, and Password & Security Management — reflecting the expanded scope of the LMS as the project matured.

A total of 8 defects were identified, of which 2 are high severity (account lockout and marks persistence), 3 are medium severity (broadcast notifications, scheduling conflict detection, and mobile login error display), and 3 are low severity. Two defects have been resolved, one is in progress, and five remain open for the next sprint.

These test cases ensure the LMS behaves according to the Software Requirements Specification (SRS) and supports reliable, secure, and role-appropriate operation for all user groups: Admin, Technical Officer, Teacher, and Student.

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Date	May 2026